Problem Statement

The objective of this analysis is to explore the Netflix dataset to identify content consumption trends, production patterns, and growth opportunities.

The goal is to help Netflix:

- Decide what type of content (Movies vs TV Shows) to produce?
- Identify high-performing genres and formats.
- Understand country-wise content distribution.
- Optimize release timing and duration.
- Expand effectively in international markets.

Basic Metrics & Dataset Overview

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load dataset
df = pd.read_csv("netflix.csv")

# Preview
df.head()
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	de
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 min	Documentaries	n
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	A p
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	T p
			Jailbirds									

Next steps: Generate code with df New interactive sheet

```
df.shape
df.info()
df.describe(include='all')
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype 
---  ------        --------------  ----- 
 0   show_id       8807 non-null   object
 1   type          8807 non-null   object
 2   title         8807 non-null   object
 3   director      6173 non-null   object
 4   cast          7982 non-null   object
 5   country       7976 non-null   object
 6   date_added    8797 non-null   object
 7   release_year  8807 non-null   int64 
 8   rating        8803 non-null   object
 9   duration      8804 non-null   object
 10  listed_in     8807 non-null   object
 11  description   8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in
count	8807	8807	8807	6173	7982	7976	8797	8807.000000	8803	8804	8807
unique	8807	2	8807	4528	7692	748	1767	NaN	17	220	514
top	s8807	Movie	Zubaan	Rajiv Chilaka	David Attenborough	United States	January 1, 2020	NaN	TV-MA	1 Season	Dramas, International Movies
freq	1	6131	1	19	19	2818	109	NaN	3207	1793	362
mean	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2014.180198	NaN	NaN	NaN
std	NaN	NaN	NaN	NaN	NaN	NaN	NaN	8.819312	NaN	NaN	NaN
min	NaN	NaN	NaN	NaN	NaN	NaN	NaN	1925.000000	NaN	NaN	NaN
25%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2013.000000	NaN	NaN	NaN
50%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2017.000000	NaN	NaN	NaN
75%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2019.000000	NaN	NaN	NaN

- The dataset contains approximately 8807 records and 12 features.
- Mix of categorical and numerical features
- Key columns:
  - type → Movie / TV Show
  - country → Production location
  - listed_in → Genre
  - duration → Minutes / Seasons
  - date_added needs conversion to datetime *Missing values present in:
  - director
  - cast
  - country
  - rating

Dataset Shape

```
df.shape
```

```
(8807, 12)
```

Observation:

- The dataset contains 8807 rows and 12 columns
- Each row represents a unique movie or TV show

Data Types of Attributes

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
```

```
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   show_id       8807 non-null   object
 1   type          8807 non-null   object
 2   title         8807 non-null   object
 3   director      6173 non-null   object
 4   cast          7982 non-null   object
 5   country       7976 non-null   object
 6   date_added    8797 non-null   object
 7   release_year  8807 non-null   int64
 8   rating        8803 non-null   object
 9   duration      8804 non-null   object
 10  listed_in     8807 non-null   object
 11  description   8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

Data Type Conversion

- **Convert Date Column**

```
df['date_added'] = pd.to_datetime(df['date_added'], errors='coerce')
df
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_i
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	2021-09-25	2020	PG-13	90 min	Documentarie
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	2021-09-24	2021	TV-MA	2 Seasons	Internationa TV Shows, T Dramas, T Mysterie
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	2021-09-24	2021	TV-MA	1 Season	Crime T Shows Internationa TV Shows, T Act..
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	2021-09-24	2021	TV-MA	1 Season	Docuseries Reality T
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	2021-09-24	2021	TV-MA	2 Seasons	Internationa TV Shows Romantic T Shows, TV ..
...	...	...	...	...	...	...	...	...	...	...	..
8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	2019-11-20	2007	R	158 min	Cult Movies Dramas Thriller
8803	s8804	TV Show	Zombie Dumb	NaN	NaN	NaN	2019-07-01	2018	TV-Y7	2 Seasons	Kids' T Korean T Shows, T Comedie
8804	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma	United States	2019-11-01	2009	R	88 min	Comedies Horror Movie

Next steps: Generate code with df New interactive sheet

- **Convert Categorical Columns**

```
cat_cols = ['type', 'rating']

for col in cat_cols:
    df[col] = df[col].astype('category')
df
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_i
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	2021-09-25	2020	PG-13	90 min	Documentarie
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...	...	...	...	...	...	...	...	...	...	...	..
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8804	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma	United States	2019-11-01	2009	R	88 min	Comedies Horror Movie

Next steps: Generate code with df New interactive sheet

Observation:

- Converting to category reduces memory usage
- Improves performance in analysis

Missing Value Detection

```
df.isnull().sum()
df
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_i
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	2021-09-25	2020	PG-13	90 min	Documentarie
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	2021-09-24	2021	TV-MA	2 Seasons	Internationa TV Shows, TV Dramas, TV Mysterie
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3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	2021-09-24	2021	TV-MA	1 Season	Docuseries Reality TV
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...	...	...	...	...	...	...	...	...	...	...	..
8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	2019-11-20	2007	R	158 min	Cult Movies Dramas Thriller
8803	s8804	TV Show	Zombie Dumb	NaN	NaN	NaN	2019-07-01	2018	TV-Y7	2 Seasons	Kids' TV Korean TV Shows, TV Comedie
8804	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma	United States	2019-11-01	2009	R	88 min	Comedies Horror Movie

Next steps: Generate code with df New interactive sheet

Observations:

- Director & Cast: High missing values
- Country: Moderate missing values
- Rating: Few missing values
- Date_added: Some missing values

Handling Missing Values

```
df['director'].fillna('Unknown', inplace=True)
df['cast'].fillna('Unknown', inplace=True)
df['country'].fillna('Unknown', inplace=True)
df['rating'].fillna(df['rating'].mode()[0], inplace=True)
df
```

```
/tmp/ipykernel_1039/883778791.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Serie
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on w

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)


  df['director'].fillna('Unknown', inplace=True)
/tmp/ipykernel_1039/883778791.py:2: FutureWarning: A value is trying to be set on a copy of a DataFrame or Serie
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on w

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)


  df['cast'].fillna('Unknown', inplace=True)
/tmp/ipykernel_1039/883778791.py:3: FutureWarning: A value is trying to be set on a copy of a DataFrame or Serie
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on w

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)


  df['country'].fillna('Unknown', inplace=True)
/tmp/ipykernel_1039/883778791.py:4: FutureWarning: A value is trying to be set on a copy of a DataFrame or Serie
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on w

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)


  df['rating'].fillna(df['rating'].mode()[0], inplace=True)
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	Unknown	United States	2021-09-25	2020	PG-13	90 min	Documentar
1	s2	TV Show	Blood & Water	Unknown	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	2021-09-24	2021	TV-MA	2 Seasons	Internatio TV Shows, Dramas, Myster
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	Unknown	2021-09-24	2021	TV-MA	1 Season	Crime Show Internatio TV Shows, Ac
3	s4	TV Show	Jailbirds New Orleans	Unknown	Unknown	Unknown	2021-09-24	2021	TV-MA	1 Season	Docuseri Reality
4	s5	TV Show	Kota Factory	Unknown	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	2021-09-24	2021	TV-MA	2 Seasons	Internatio TV Show Romantic Shows, TV
...	...	...	...	...	...	...	...	...	...	...	
8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	2019-11-20	2007	R	158 min	Cult Movi Dram Thrill
8803	s8804	TV Show	Zombie Dumb	Unknown	Unknown	Unknown	2019-07-01	2018	TV-Y7	2 Seasons	Kids' T Korean Shows, Comed
8804	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	2019-11-01	2009	R	88 min	Comedi Horror Mov
8805	s8806	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy	United States	2020-01-11	2006	PG	88 min	Childrer Family Movi

Next steps: Generate code with df New interactive sheet

Statistical Summary

```
df.describe(include='all')
df
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	Unknown	United States	2021-09-25	2020	PG-13	90 min	Documentaries
1	s2	TV Show	Blood & Water	Unknown	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	Unknown	2021-09-24	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...
3	s4	TV Show	Jailbirds New Orleans	Unknown	Unknown	Unknown	2021-09-24	2021	TV-MA	1 Season	Docuseries, Reality TV
4	s5	TV Show	Kota Factory	Unknown	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...
...	...	...	...	...	...	...	...	...	...	...	...
8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	2019-11-20	2007	R	158 min	Cult Movies, Dramas, Thrillers
8803	s8804	TV Show	Zombie Dumb	Unknown	Unknown	Unknown	2019-07-01	2018	TV-Y7	2 Seasons	Kids' TV, Korean TV Shows, TV Comedies
8804	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma	United States	2019-11-01	2009	R	88 min	Comedies, Horror Movies

Next steps: Generate code with df New interactive sheet

Key Observations: Based on Release Year

- Wide range of years
- Majority content is post-2000

Duration

```
df[['duration_num', 'duration_unit']] = df['duration'].str.split(' ', n=1, expand=True)
df['duration_num'] = df['duration_num'].astype(float)


df_movies = df[df['type'] == 'Movie'].copy()
df_tv = df[df['type'] == 'TV Show'].copy()


df_movies['duration_timedelta'] = pd.to_timedelta(df_movies['duration_num'], unit='m')
```

```
df_movies['duration_days'] = df_movies['duration_timedelta'].dt.days


df_tv['duration_timedelta'] = pd.to_timedelta(df_tv['duration_num'] * 365, unit='d')
df_tv['duration_days'] = df_tv['duration_timedelta'].dt.days


print(df_movies['duration_days'].describe())
print(df_tv['duration_days'].describe())


df['date_added'] = pd.to_datetime(df['date_added'], errors='coerce')
df['year_added'] = df['date_added'].dt.year
df['month_added'] = df['date_added'].dt.month
df['day_added'] = df['date_added'].dt.day
```

```
count    6128.0
mean        0.0
std         0.0
min         0.0
25%         0.0
50%         0.0
75%         0.0
max         0.0
Name: duration_days, dtype: float64
count    2676.000000
mean      644.205904
std       577.704445
min       365.000000
25%       365.000000
50%       365.000000
75%       730.000000
max      6205.000000
Name: duration_days, dtype: float64
```

Observation

1. Movie Duration

Insights:

- Movies follow a highly standardized duration pattern
- Industry norm (90 to 120 mins) is clearly visible
- Very short or very long movies are rare

2. TV Show Duration

Insights:

- Most shows have 1-2 seasons
- Only a few shows run for many seasons
- Indicates focus on short-format, binge content

Recommendation:

- Invest more in mini-series and limited series(1-3 seasons)
- Avoid long-running shows unless highly successful

3. Year-wise Trends (year_added)

Insights:

1. Reflects Netflix global expansion phase
2. Strong shift toward original and international content

Double-click (or enter) to edit

Recommendation:

- Maintain high content production but focus on quality over quantity

4. Month-wise Trends (month_added)

Insights:

- Netflix follows seasonal release strategy
- Aligns with holiday season and high user engagement periods

Recommendation:

- Launch major releases in festive seasons and Year-end windows

5. Day-wise Trends (day_added)

Insights:

- Day-level variation is not strategically significant
- Monthly/seasonal trends matter more

Recommendation:

- Focus on month/quarter-level planning, not daily timing

So Netflix optimizes movies for consistent runtime while focusing on short, binge-worthy TV shows, with content releases strategically aligned to seasonal demand and global expansion trends.

3. Non-Graphical Analysis: Value Counts & Unique Attributes

Double-click (or enter) to edit

1. Value Counts Analysis

- Content Type Distribution

```
df['type'].value_counts()
df
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	Unknown	United States	2021-09-25	2020	PG-13	90 min	Documentar
1	s2	TV Show	Blood & Water	Unknown	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	2021-09-24	2021	TV-MA	2 Seasons	Internatio TV Shows, Dramas, Myster
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	Unknown	2021-09-24	2021	TV-MA	1 Season	Crime Shov Internatio TV Shows, Ac
3	s4	TV Show	Jailbirds New Orleans	Unknown	Unknown	Unknown	2021-09-24	2021	TV-MA	1 Season	Docuseri Reality
4	s5	TV Show	Kota Factory	Unknown	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	2021-09-24	2021	TV-MA	2 Seasons	Internatio TV Shov Romantic Shows, TV
...	...	...	...	...	...	...	...	...	...	...	
8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	2019-11-20	2007	R	158 min	Cult Movi Dram Thrill
8803	s8804	TV Show	Zombie Dumb	Unknown	Unknown	Unknown	2019-07-01	2018	TV-Y7	2 Seasons	Kids' T Korean Shows, Comed
8804	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	2019-11-01	2009	R	88 min	Comedi Horror Mov
8805	s8806	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	2020-01-11	2006	PG	88 min	Childrer Family Movi Comed
8806	s8807	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	2019-03-02	2015	TV-14	111 min	Dram Internatio Movies, Mu & Music

8807 rows × 17 columns

Next steps: Generate code with df New interactive sheet

Insights:

- Netflix catalog is movie-dominated, but TV Shows are growing

Top Countries

```
df['country'].value_counts().head(10)
```

country	count
United States	2818
India	972
Unknown	831
United Kingdom	419
Japan	245
South Korea	199
Canada	181
Spain	145
France	124
Mexico	110

dtype: int64

Insight:

- USA ranks highest.India, UK,Japan,South Korea follows
- Content production is highly concentrated in few countries

Top Genres

```
df['listed_in'].value_counts().head(10)
```

listed_in	count
Dramas, International Movies	362
Documentaries	359
Stand-Up Comedy	334
Comedies, Dramas, International Movies	274
Dramas, Independent Movies, International Movies	252
Kids' TV	220
Children & Family Movies	215
Children & Family Movies, Comedies	201
Documentaries, International Movies	186
Dramas, International Movies, Romantic Movies	180

dtype: int64

Insight:

- Drama is most frequent
- International content appears prominently
- Netflix focuses on globally appealing genres

Ratings Distribution

```
df['rating'].value_counts()
df
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	Unknown	United States	2021-09-25	2020	PG-13	90 min	Documentar
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8807 rows × 17 columns

Next steps: Generate code with df New interactive sheet

Insights:

- TV-MA and TV-14 dominate
- Kids content is limited
- Platform targets adult and young adult audience

Top Actors

```
df['cast'].value_counts().head(10)
df
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	Unknown	United States	2021-09-25	2020	PG-13	90 min	Documentar
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3	s4	TV Show	Jailbirds New Orleans	Unknown	Unknown	Unknown	2021-09-24	2021	TV-MA	1 Season	Docuseri Reality
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8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	2019-11-20	2007	R	158 min	Cult Movi Dram Thrill
8803	s8804	TV Show	Zombie Dumb	Unknown	Unknown	Unknown	2019-07-01	2018	TV-Y7	2 Seasons	Kids' T Korean Shows, Comed
8804	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	2019-11-01	2009	R	88 min	Comedi Horror Mov
8805	s8806	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	2020-01-11	2006	PG	88 min	Childrer Family Movi Comed
8806	s8807	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	2019-03-02	2015	TV-14	111 min	Dram Internatio Movies, Mu & Music

8807 rows × 17 columns

Next steps: Generate code with df New interactive sheet

Insights:

- Few actors appear frequently
- Most actors appear only once
- Content is highly diverse in casting

Top Directors

```
df['director'].value_counts().head(10)
df
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	Unknown	United States	2021-09-25	2020	PG-13	90 min	Documentar
1	s2	TV Show	Blood & Water	Unknown	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	2021-09-24	2021	TV-MA	2 Seasons	Internatio TV Shows, Dramas, Myster
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	Unknown	2021-09-24	2021	TV-MA	1 Season	Crime Shov Internatio TV Shows, Ac
3	s4	TV Show	Jailbirds New Orleans	Unknown	Unknown	Unknown	2021-09-24	2021	TV-MA	1 Season	Docuseri Reality
4	s5	TV Show	Kota Factory	Unknown	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	2021-09-24	2021	TV-MA	2 Seasons	Internatio TV Shov Romantic Shows, TV
...	...	...	...	...	...	...	...	...	...	...	
8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	2019-11-20	2007	R	158 min	Cult Movi Dram Thrill
8803	s8804	TV Show	Zombie Dumb	Unknown	Unknown	Unknown	2019-07-01	2018	TV-Y7	2 Seasons	Kids' Korean Shows, Comed
8804	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	2019-11-01	2009	R	88 min	Comedi Horror Mov
8805	s8806	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	2020-01-11	2006	PG	88 min	Childrer Family Movi Comed
8806	s8807	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	2019-03-02	2015	TV-14	111 min	Dram Internatio Movies, Mu & Music

8807 rows × 17 columns

Next steps: Generate code with df New interactive sheet

Insights:

- No single dominant director
- Netflix relies on diverse creators, not specific individuals

2. Unique Attributes Analysis

Unique Count of Key Columns

```
df.nunique()
df
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	Unknown	United States	2021-09-25	2020	PG-13	90 min	Documentar
1	s2	TV Show	Blood & Water	Unknown	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	2021-09-24	2021	TV-MA	2 Seasons	Internatio TV Shows, Dramas, Myster
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	Unknown	2021-09-24	2021	TV-MA	1 Season	Crime Shov Internatio TV Shows, Ac
3	s4	TV Show	Jailbirds New Orleans	Unknown	Unknown	Unknown	2021-09-24	2021	TV-MA	1 Season	Docuseri Reality
4	s5	TV Show	Kota Factory	Unknown	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	2021-09-24	2021	TV-MA	2 Seasons	Internatio TV Shov Romantic Shows, TV
...	...	...	...	...	...	...	...	...	...	...	
8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	2019-11-20	2007	R	158 min	Cult Movi Dram Thrill
8803	s8804	TV Show	Zombie Dumb	Unknown	Unknown	Unknown	2019-07-01	2018	TV-Y7	2 Seasons	Kids' Korean Shows, Comed
8804	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	2019-11-01	2009	R	88 min	Comedi Horror Mov
8805	s8806	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	2020-01-11	2006	PG	88 min	Childrer Family Movi Comed
8806	s8807	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	2019-03-02	2015	TV-14	111 min	Dram Internatio Movies, Mu & Music

8807 rows × 17 columns

Next steps: Generate code with df New interactive sheet

Insights:

- Large and diverse content library with high uniqueness in titles
- Limited unique countries
- Multi-genre classification improves content discoverability
- Standard ratings help in audience segmentation
- Netflix's catalog is highly diverse in content and creators but concentrated geographically, with strong dominance of movies and mature audience targeting.

So we can say :

- Netflix content is diverse but geographically concentrated
- Movies dominate, but TV Shows are growing rapidly
- Genres are multi-tagged, enhancing recommendation systems
- Audience targeting is skewed toward mature viewers
- Content creators (actors/directors) are highly distributed

Recommendation:

- Content Strategy : Increase TV Show production for engagement
- Expansion Strategy : Expand into underrepresented countries
- Genre Strategy : Focus on Drama and International content
- Audience Strategy : Increase kids and family content

4. Visual Analysis (Univariate & Bivariate)

Preprocessing of Data

```
df = df.copy()

cols = ['cast', 'director', 'country', 'listed_in']

for col in cols:
    df[col] = df[col].str.split(',')

df = df.explode('cast')
df = df.explode('director')
df = df.explode('country')
df = df.explode('listed_in')


for col in cols:
    df[col] = df[col].str.strip()
df
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	Unknown	United States	2021-09-25	2020	PG-13	90 min	Documentaries
1	s2	TV Show	Blood & Water	Unknown	Ama Qamata	South Africa	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows
1	s2	TV Show	Blood & Water	Unknown	Ama Qamata	South Africa	2021-09-24	2021	TV-MA	2 Seasons	TV Dramas
1	s2	TV Show	Blood & Water	Unknown	Ama Qamata	South Africa	2021-09-24	2021	TV-MA	2 Seasons	TV Mysteries
1	s2	TV Show	Blood & Water	Unknown	Khosi Ngema	South Africa	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows
...	...	...	...	...	...	...	...	...	...	...	...
8806	s8807	Movie	Zubaan	Mozez Singh	Anita Shabdish	India	2019-03-02	2015	TV-14	111 min	International Movies
8806	s8807	Movie	Zubaan	Mozez Singh	Anita Shabdish	India	2019-03-02	2015	TV-14	111 min	Music & Musicals
8806	s8807	Movie	Zubaan	Mozez Singh	Chittaranjan Tripathy	India	2019-03-02	2015	TV-14	111 min	Dramas
8806	s8807	Movie	Zubaan	Mozez Singh	Chittaranjan Tripathy	India	2019-03-02	2015	TV-14	111 min	International Movies
8806	s8807	Movie	Zubaan	Mozez Singh	Chittaranjan Tripathy	India	2019-03-02	2015	TV-14	111 min	Music & Musicals

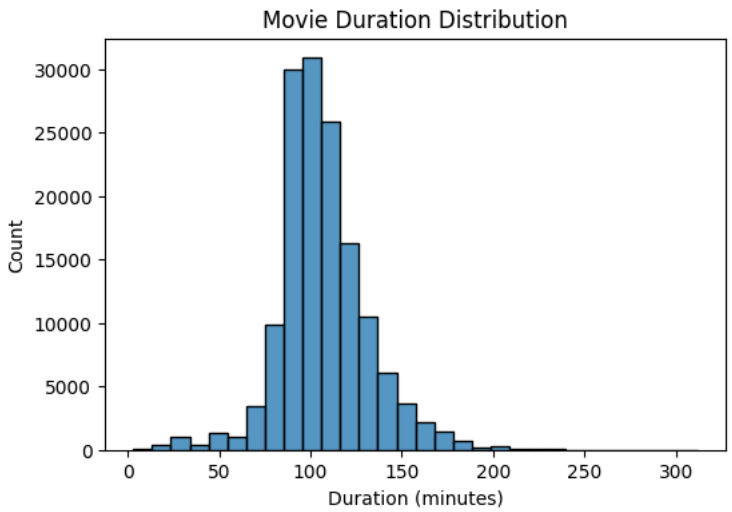
202065 rows × 17 columns

1. Univariate Analysis (Continuous Variables)

1.1 Movie Duration Distribution (Histogram)

```
print(df.columns)
df[['duration_num', 'duration_unit']] = df['duration'].str.split(' ', n=1, expand=True)
df['duration_int'] = pd.to_numeric(df['duration_num'], errors='coerce')
df = df[df['duration_int'].notna()].copy()
plt.figure(figsize=(6,4))

sns.histplot(df[df['type']=='Movie']['duration_int'], bins=30)

plt.title("Movie Duration Distribution")
plt.xlabel("Duration (minutes)")

plt.show()
```

```
Index(['show_id', 'type', 'title', 'director', 'cast', 'country', 'date_added',
       'release_year', 'rating', 'duration', 'listed_in', 'description',
       'duration_num', 'duration_unit', 'year_added', 'month_added',
       'day_added', 'duration_int'],
      dtype='object')
```

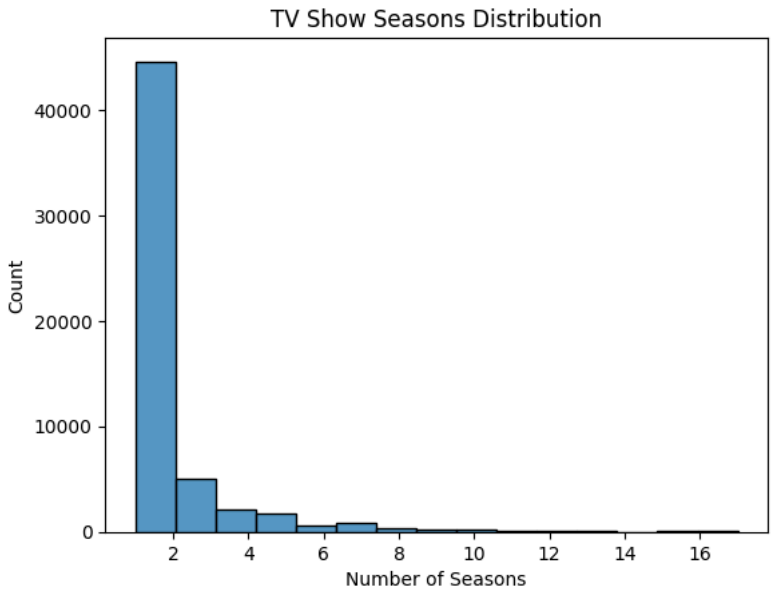


Insights:

- Most movies fall between 90-120 minutes
- Extreme values are rare
- Distribution is approximately normal

1.2. TV Shows Seasons Distribution

```
sns.histplot(df[df['type']=='TV Show']['duration_int'], bins=15)
plt.title("TV Show Seasons Distribution")
plt.xlabel("Number of Seasons")
plt.show()
```



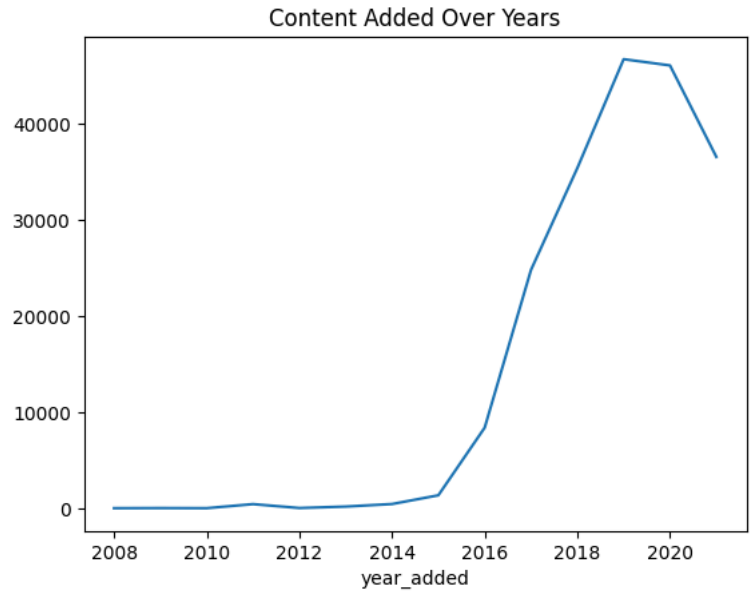
Insights:

- Majority shows have 1-2 seasons
- Highly right-skewed distribution
- Very few long-running shows

1.3. Content Added Over Years

```
df['year_added'].value_counts().sort_index().plot(kind='line')
plt.title("Content Added Over Years")
```

```
plt.title("Content Added Over Years")
plt.show()
```



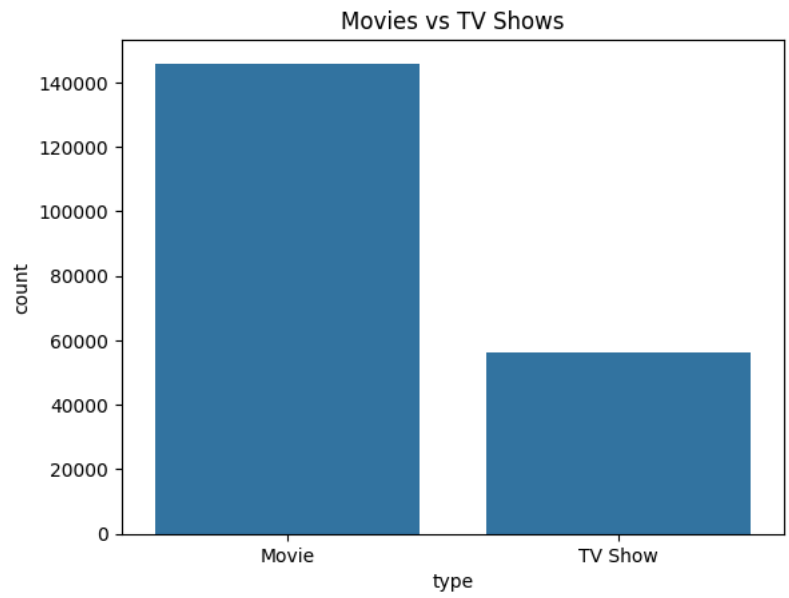
Insights:

- Rapid growth after 2015
- Peak during 2017-2020

2. Univariate Analysis (Categorical Variables)

2.1 Movies vs TV Shows

```
sns.countplot(x='type', data=df)
plt.title("Movies vs TV Shows")
plt.show()
```



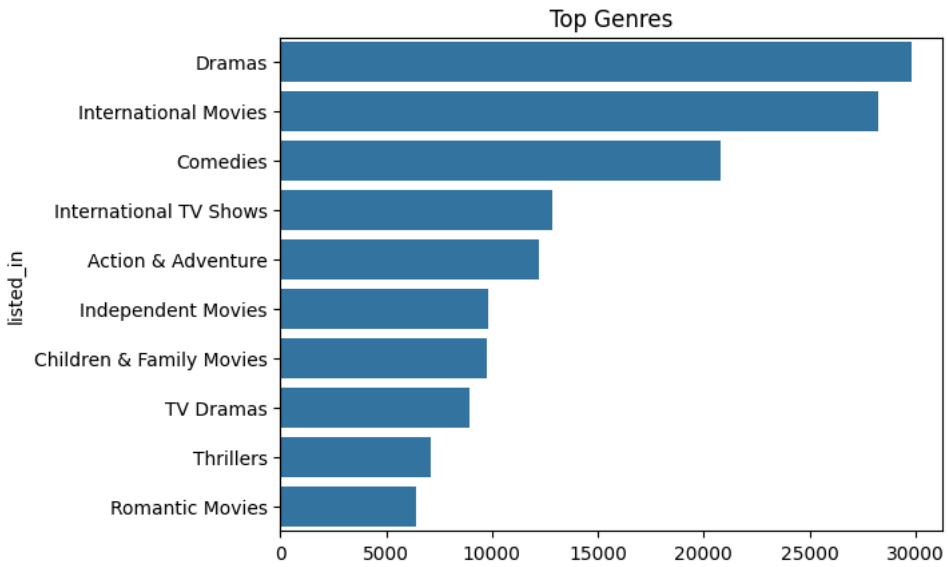
Insights:

- Movies dominate but gap is shrinking

2.2 Top Genres

```
top_genres = df['listed_in'].value_counts().head(10)
```

```
sns.barplot(x=top_genres.values, y=top_genres.index)
plt.title("Top Genres")
plt.show()
```

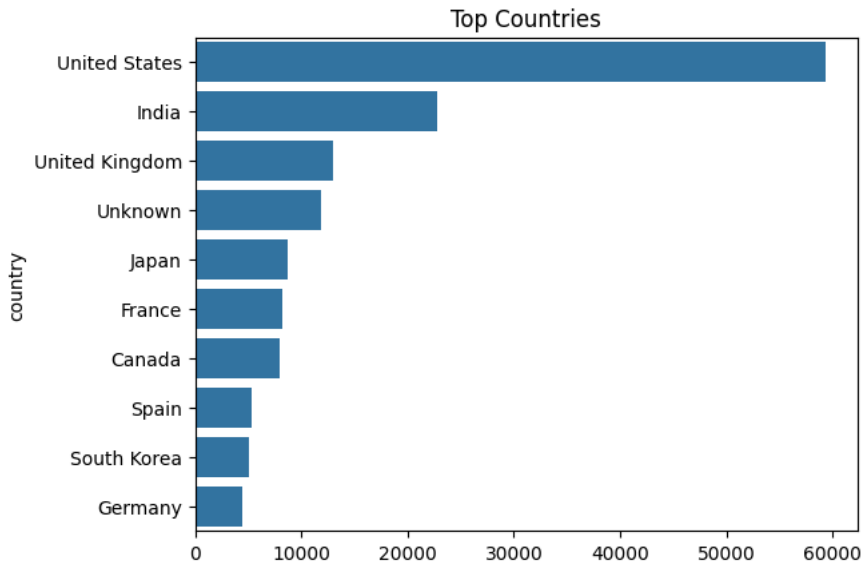


Insights:

- Drama and International movies dominate
- Romantic Movies is the least preferred genre

2.3 Top Countries

```
top_countries = df['country'].value_counts().head(10)

sns.barplot(x=top_countries.values, y=top_countries.index)
plt.title("Top Countries")
plt.show()
```



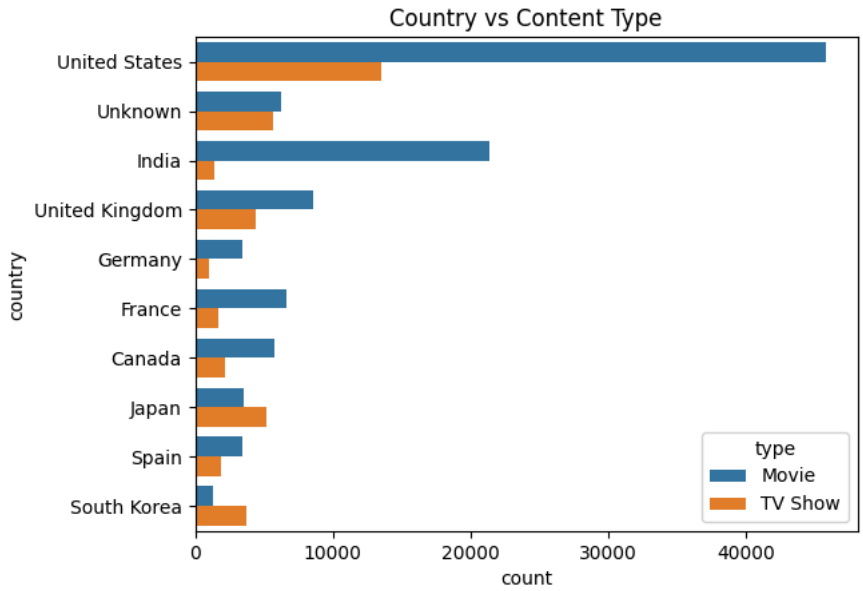
Insight:

- USA dominates,followed by India,UK,Japan.
- India growing(2nd placed)
- Germany is the bottom country

3. Bivariate Analysis

3.1 Country vs Content Type

```
sns.countplot(y='country', hue='type',
              data=df[df['country'].isin(top_countries.index)])
plt.title("Country vs Content Type")
plt.show()
```

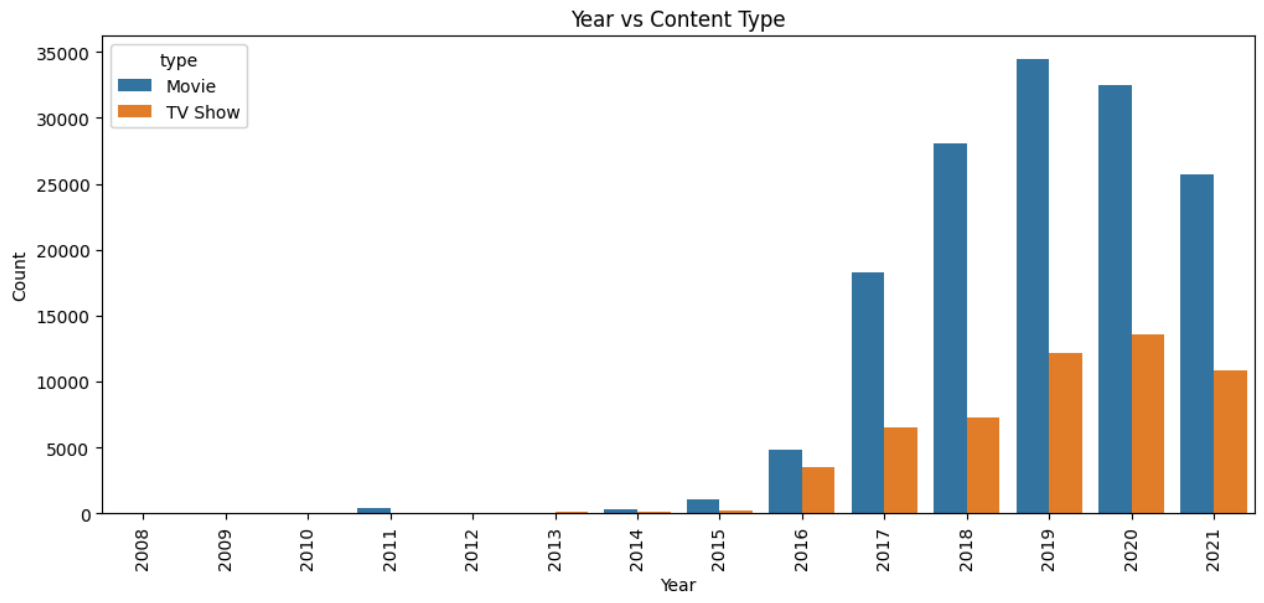


Insights:

- India and US : Movie dominant
- South Korea and Japan : TV Shows dominant

3.2 Year vs Content Type

```
df['date_added'] = pd.to_datetime(df['date_added'], errors='coerce')

# Extract year
df['year_added'] = df['date_added'].dt.year

# Drop missing values
df = df.dropna(subset=['year_added'])

# Convert to integer
df['year_added'] = df['year_added'].astype(int)

plt.figure(figsize=(12,5))

sns.countplot(x='year_added', hue='type', data=df)

plt.xticks(rotation=90)
plt.title("Year vs Content Type")
plt.xlabel("Year")
plt.ylabel("Count")

plt.show()
```

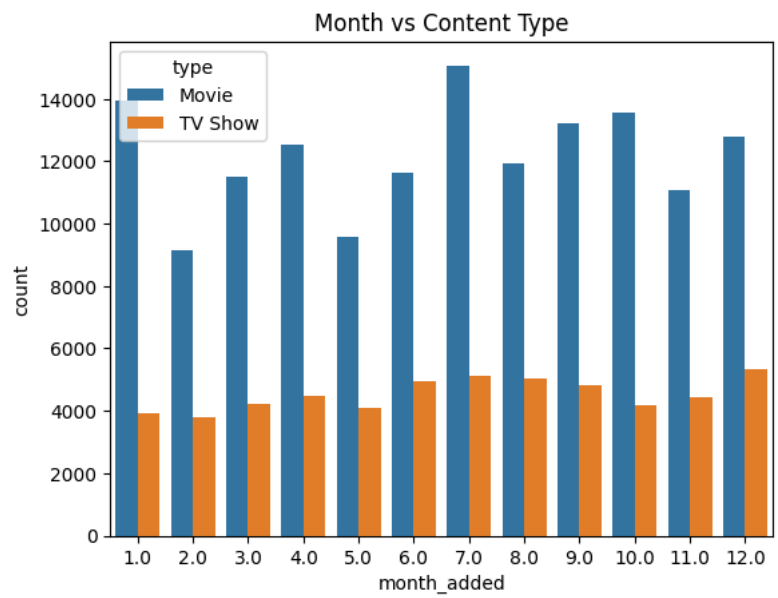


Insights:

- TV Shows increasing rapidly in recent years
- Clear shift toward series content
- 2019 is the year of maximum count of movies

3.3 Month vs Content Type

```
sns.countplot(x='month_added', hue='type', data=df)
plt.title("Month vs Content Type")
plt.show()
```



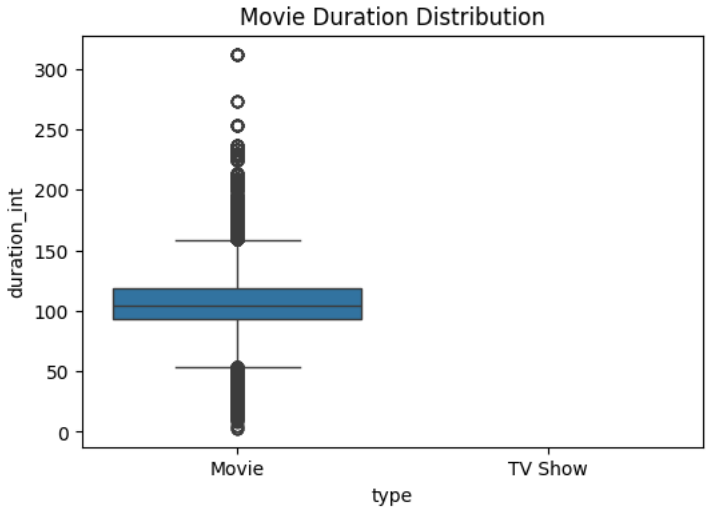
Insights:

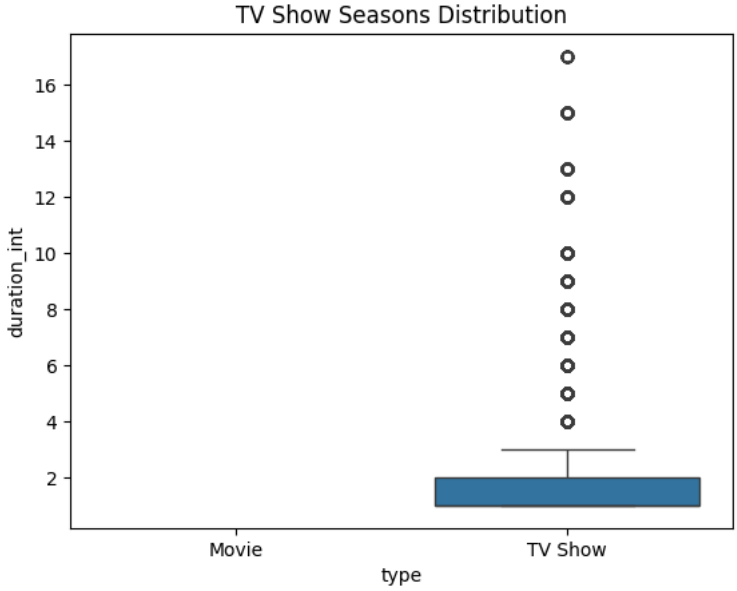
- Peak releases in Q3-Q4
- Seasonal strategy visible

3.4 Rating vs Content Type (Boxplot)

```
# Split duration into number + unit
df[['duration_num', 'duration_unit']] = df['duration'].str.split(' ', n=1, expand=True)
```

```
# Convert to numeric safely
df['duration_int'] = pd.to_numeric(df['duration_num'], errors='coerce')
df['duration_int'].isnull().sum()
df = df[df['duration_int'].notna()].copy()
df['duration_int'] = df['duration_int'].astype(float)
df['type'] = df['type'].astype('category')
plt.figure(figsize=(6,4))

sns.boxplot(x='type', y='duration_int', data=df[df['type']=='Movie'])
plt.title("Movie Duration Distribution")
plt.show()

sns.boxplot(x='type', y='duration_int', data=df[df['type']=='TV Show'])
plt.title("TV Show Seasons Distribution")
plt.show()
```





Insights:

- Movies show tight distribution (90 to 120 mins)
- TV Shows show wide variation (1 to 17 seasons)
- Movies are standardized
- TV Shows are more flexible and variable

Recommendation:

- Keep movies within 90 to 120 mins
- Focus on short TV series (1 to 3 seasons)

So Netflix visual trends highlight a shift toward short, binge-worthy TV content, supported by strong international expansion and seasonal release strategies.Movie duration is standardized, TV Shows are variable but short Netflix is shifting from movies → TV Shows Content production is region-specific Growth is post-2015 driven Release strategy is seasonal Genres and countries show clear concentration patterns

Missing Value & Outlier Check

1.Check Missing Values

```
df.isnull().sum()
df
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	Unknown	United States	2021-09-25	2020	PG-13	90 min	Documentaries
1	s2	TV Show	Blood & Water	Unknown	Ama Qamata	South Africa	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows
1	s2	TV Show	Blood & Water	Unknown	Ama Qamata	South Africa	2021-09-24	2021	TV-MA	2 Seasons	TV Dramas
1	s2	TV Show	Blood & Water	Unknown	Ama Qamata	South Africa	2021-09-24	2021	TV-MA	2 Seasons	TV Mysteries
1	s2	TV Show	Blood & Water	Unknown	Khosi Ngema	South Africa	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows
...	...	...	...	...	...	...	...	...	...	...	...
8806	s8807	Movie	Zubaan	Mozez Singh	Anita Shabdish	India	2019-03-02	2015	TV-14	111 min	International Movies
8806	s8807	Movie	Zubaan	Mozez Singh	Anita Shabdish	India	2019-03-02	2015	TV-14	111 min	Music & Musicals
8806	s8807	Movie	Zubaan	Mozez Singh	Chittaranjan Tripathy	India	2019-03-02	2015	TV-14	111 min	Dramas
8806	s8807	Movie	Zubaan	Mozez Singh	Chittaranjan Tripathy	India	2019-03-02	2015	TV-14	111 min	International Movies
8806	s8807	Movie	Zubaan	Mozez Singh	Chittaranjan Tripathy	India	2019-03-02	2015	TV-14	111 min	Music & Musicals

200316 rows × 18 columns

Observations:

- Director & Cast :- High number of missing values
- Country :- Moderate missing values
- Rating :- Few missing values
- Date_added :- Some missing values

Insights:

- Missing values are mostly in categorical columns
- These do not represent absence of data but rather unavailable information

- Numerical columns are mostly complete

Treatment Applied

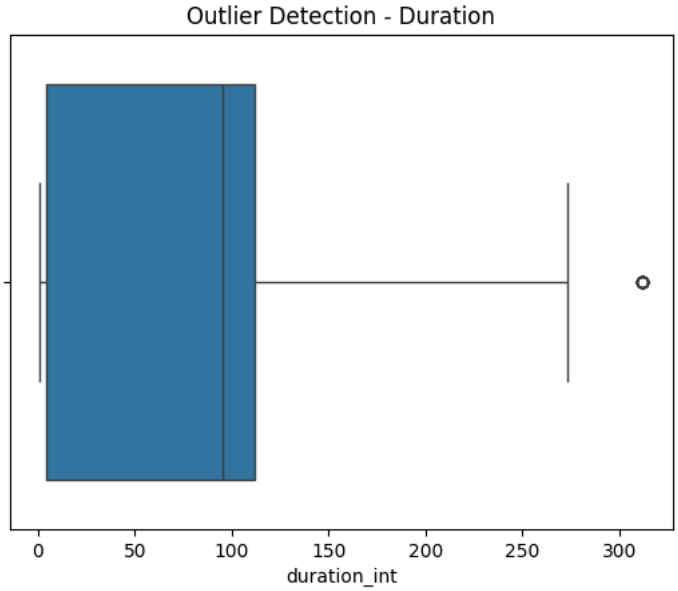
```
df['director'] = df['director'].fillna('Unknown')
df['cast'] = df['cast'].fillna('Unknown')
df['country'] = df['country'].fillna('Unknown')
df['rating'] = df['rating'].fillna(df['rating'].mode()[0])
df
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	Unknown	United States	2021-09-25	2020	PG-13	90 min	Documentaries
1	s2	TV Show	Blood & Water	Unknown	Ama Qamata	South Africa	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows
1	s2	TV Show	Blood & Water	Unknown	Ama Qamata	South Africa	2021-09-24	2021	TV-MA	2 Seasons	TV Dramas
1	s2	TV Show	Blood & Water	Unknown	Ama Qamata	South Africa	2021-09-24	2021	TV-MA	2 Seasons	TV Mysteries
1	s2	TV Show	Blood & Water	Unknown	Khosi Ngema	South Africa	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows
...	...	...	...	...	...	...	...	...	...	...	...
8806	s8807	Movie	Zubaan	Mozez Singh	Anita Shabdish	India	2019-03-02	2015	TV-14	111 min	International Movies
8806	s8807	Movie	Zubaan	Mozez Singh	Anita Shabdish	India	2019-03-02	2015	TV-14	111 min	Music & Musicals
8806	s8807	Movie	Zubaan	Mozez Singh	Chittaranjan Tripathy	India	2019-03-02	2015	TV-14	111 min	Dramas
8806	s8807	Movie	Zubaan	Mozez Singh	Chittaranjan Tripathy	India	2019-03-02	2015	TV-14	111 min	International Movies
8806	s8807	Movie	Zubaan	Mozez Singh	Chittaranjan Tripathy	India	2019-03-02	2015	TV-14	111 min	Music & Musicals

200316 rows × 18 columns

Boxplot for Outlier Detection

```
sns.boxplot(x=df['duration_int'])
plt.title("Outlier Detection - Duration")
plt.show()
```



Insights:

- Outliers exist but are real-world valid cases
- These represent: Special content (short films, documentaries), Long-running popular series
- Most movies between 60 to 120 minutes
- Most shows have 1 to 2 seasons
- The dataset has manageable missing values and minimal impactful outliers, ensuring reliable analysis without significant data loss.

Insights Based on Non-Graphical & Visual Analysis

1.Comments on Range of Attributes:

- Release Year: Ranges from 1925 to 2021, but majority of content is post-2000
- Movie Duration: Ranges from very short (3 mins) to very long (300+ mins), but most movies lie between 90 to 120 minutes
- TV Shows Duration (Seasons): Range: 1 to 17 seasons, with most shows having 1 to 2 seasons
- Country Distribution: Content spans across 80+ countries, but ~80% comes from top 10 countries
- Ratings: Limited categories, dominated by TV-MA and TV-14

Insights:

- Netflix content is:
- Modern and recent
- Standardized in format (duration)
- Geographically concentrated

2.Distribution & Relationships

Double-click (or enter) to edit

2.1 Movie Duration Distribution

- Shape: Approximately normal (slightly right-skewed)
- Most values cluster around 90–100 minutes

Insights:

- Indicates industry standardization
- Optimized for viewer engagement and completion

2.2 TV Shows Distribution

- Shape: Highly right-skewed
- Majority: 1 to 2 seasons

Insights:

- Reflects Netflix strategy of: Short, bingeable content
- Lower risk compared to long-running shows

2.3 Movies vs TV Shows

- Ratio ≈ 2.3:1 (Movies > TV Shows)
- Gap is decreasing over time

Insights:

- Netflix is shifting toward TV Shows for retention
- Movies for initial engagement

2.4 Content Growth Over Time

- Significant rise after 2015
- Peak growth during 2017 to 2020

Insights:

- Reflects global expansion strategy
- Increased focus on original content

2.5 Ratings Distribution

2.5.1 Univariate Plots

Movies vs TV Shows (Countplot)

- Movies dominate
- TV Shows increasing

Insight:

- Shift toward series-based content

Duration Histogram

- Movies : Normal distribution
- TV Shows : Right-skewed

Insights:

- Movies are standardized
- TV Shows are flexible

Content Over Years (Line Plot)

- Sharp rise post-2015

Insights:

- Growth driven by global expansion

Top Genres (Barplot)

- Drama & International dominate

Insights:

- Focus on broad and global appeal

Top Countries (Barplot)

- USA leads significantly

Insight:

- High geographic concentration

2.5.2 Bivariate Plots

Country vs Content Type

- India/USA :- Movies
- South Korea :- TV Shows

Insight:

- Reflects regional storytelling preferences

Year vs Content Type

- TV Shows increasing faster

Insight:

- Netflix shifting toward retention-driven content

Month vs Content Type

Insights:

- Peak in Q3 to Q4
- Strategic seasonal releases

Type vs Duration (Boxplot)

- Movies :- consistent range
- TV Shows :- wide variation

Insights:

- Movies optimized for consistency
- TV Shows vary based on storytelling

Final Insights:

- Netflix is transitioning from a movie-heavy platform to a TV show-focused platform
- Content is: Short, bingeable, and globally distributed
- Growth is driven by: International expansion and Localized content production
- Strong reliance on: Few countries and Mature audience segment

Netflix's strategy is shifting toward short, binge-worthy TV content supported by global expansion and localized storytelling, while maintaining standardized movie formats for mass appeal.

Business Insights:

Key Business Insights from Analysis:

1.Shift from Movies to TV Shows

Observation:

- Movies dominate overall content (2.3:1 ratio)
- However, TV Shows have grown rapidly in recent years

Business Insight:

- Netflix is transitioning toward a retention-driven model
- TV Shows keep users engaged longer than movies

2. Strong Growth of International Content

Observation:

- Significant content from:India, South Korea and UK
- International genre appears frequently

Business Insight:

- Growth is driven by localized storytelling
- Global audiences prefer region-specific content

3. Dominance of Drama & Multi-Genre Content

Observation:

- Drama is the most common genre
- Content often tagged with multiple genres

Business Insight:

- Drama has universal appeal
- Multi-genre tagging improves: Discoverability and Recommendation systems

4. Standardization of Movie Duration

Observation:

- Most movies fall within 90–120 minutes

Business Insight:

- Content is optimized for: Viewer attention span and Completion rates

5. Preference for Short TV Series

Observation:

- Majority shows have 1 to 2 seasons
- Very few long-running series

Business Insight:

- Netflix prefers:Low-risk content investment and
- Binge-worthy formats

6. Rapid Content Expansion (Post-2015)

Observation:

- Massive increase in content additions after 2015

Business Insight:

- Reflects: Global expansion strategy and
- Heavy investment in original content

7. Adult Audience Dominance

Observation:

- Majority content rated: TV-MA TV-14

Business Insight:

- Netflix is primarily targeting: Young adults & mature viewers
- Kids/family segment is underutilized

8. Geographic Concentration Risk

Observation:

- Majority content from: USA,India,UK

Business Insight:

- Over-dependence on few countries
- Risk of: Limited cultural diversity and Market saturation

9. Seasonal Content Release Strategy

Observation:

- Content peaks in Q3 & Q4 (Oct–Dec)

Business Insight:

- Netflix aligns releases with: Holidays and High engagement periods

10. Highly Diverse Talent Pool

Observation:

- No dominant actors/directors
- Long-tail distribution

Business Insight:

- Netflix strategy: Promote diverse creators and
- Avoid dependency on specific individuals

Final Business Understanding:

- Netflix operates on a global, data-driven content strategy
- Focus areas:
- Retention (TV Shows)
- Localization (International content)
- Efficiency (short formats)

Summary:

Netflix's growth is driven by a strategic shift toward binge-worthy TV content, strong investment in international markets, and optimized content formats tailored for engagement and global scalability.

Recommendations (Actionable Business Steps):

1. Focus More on TV Shows

- Increase production of TV Shows and web series

- Create short series (1-3 seasons) that are easy to binge

Because : TV Shows keep users engaged longer than movies

2. Expand International Content

- Invest more in content from: India,South Korea,Europe
- Produce shows in local languages

Because : Local content attracts more regional subscribers

3. Prioritize Popular Genres

- Focus on: Drama,Crime,Thriller
- Continue producing multi-genre content

Because: These genres have the highest demand globally

4. Increase Kids & Family Content

- Create more: Animated shows and Family-friendly movies

Because : Current content is heavily focused on adults, missing a large audience segment

5. Maintain Ideal Movie Duration

- Keep most movies within 90-120 minutes

Because : This duration is preferred by viewers and improves completion rates

6. Invest in Short, Binge-Worthy Series

- Focus on: Mini-series and Limited episodes

Because : Short content is easier to consume and highly engaging

7. Optimize Release Timing

- Release major content during: Festive seasons and Year-end (Oct-Dec)

Because : Higher user activity during these periods

8. Reduce Dependence on Few Countries

- Expand production in: Africa,Southeast Asia,Middle East
- List item

Because : Reduces risk and increases global reach

9. Support Diverse Creators

- Work with: New directors and Regional actors

Because : Brings fresh content and wider audience appeal

10. Balance Content Strategy

- Maintain a mix of: Movies - for quick engagement and
- TV Shows - for long-term retention

Final Recommendation

Netflix should focus on creating short, engaging TV content, expand internationally with localized shows, and diversify its audience by including more family-friendly content while maintaining strong release timing and content balance.

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